University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Mid Term Examination, Autumn 2025
Course Title: Database Management System
Course Code: CSE 315

Marks: 20 **Time: 1*(one)* hour**

(Answer all questions)

Q.No.	Questions	Marks
1.	a) A university database stores student information, including student ID, email, phone number, and course enrollment details. Identify and explain the following types of database keys with their specific purposes: 1. Primary Key 2. Foreign Key 3. Candidate Key 4. Super Key	[06]
	b) A student is registering for courses through the university portal. The system starts the registration process, updates the student's course list, and then suddenly faces a network failure before saving the changes permanently. Based on the states of a transaction, identify the different states this registration process goes through. Briefly explain what state the transaction will end up in due to the failure.	[04]
2)	a) Consider the following relation: • Employee(EmpID, Name, Dept, Salary) • Department(DeptID, DeptName, Location) Write relational algebra queries to: $\sigma_{sal} > 50000$ a) Retrieve employees who earn more than 50,000. b) List names of employees who work in the 'HR' department	[04]
	b) TravelBooking.com operates globally with users from different continents. They're experiencing latency issues and are considering whether to upgrade their current 2-tier architecture to 3-tier, or implement a distributed database system. Assess the pros and cons of each architectural approach for their global operations. Judge which solution would be most cost-effective while maintaining performance. Justify your recommendation with specific criteria related to scalability, maintenance, and user experience	[06]